

Signal Processing: Methods and Algorithms

Laboratory 5

1 Analysis of atomic clock data

The file `AtomicClock.txt` contains a signal $X[n]$ which represents the frequency deviation of an atomic clock undergoing tests for space flight certification. Do the following:

1. Estimate the mean value $\hat{\mu}_X[n]$ of $X[n]$ by using the function `sliding_mean` written in the second Laboratory. Modify the function by adding the additional input parameter `Istep`

$$[\text{mu_est}] = \text{sliding_mean}(\mathbf{x}, \text{Nw}, \text{Istep})$$

The parameter `Istep` is the time interval between two consecutive time instants at which the sliding mean must be evaluated (use a large `Istep` to reduce the computational time).

2. Plot $X[n]$ and the estimated mean value $\hat{\mu}_X[n]$ in the same figure.
3. Estimate the standard deviation $\hat{\sigma}_X[n]$ of $X[n]$ by writing and using the function

$$[\text{std_est}] = \text{sliding_std}(\mathbf{x}, \text{Nw}, \text{Istep})$$

which estimates the sliding standard deviation of $X[n]$. **Hint:** The fastest way to write this function is by modifying the `sliding_mean` function.

4. Plot $\hat{\sigma}_X[n]$. Is the standard deviation constant with time? Are the variations due to the random fluctuations of the estimate, or is there something more?
5. In the same figure, plot $X[n]$, the estimated mean value $\hat{\mu}_X[n]$, and the signals

$$s_1[n] = \hat{\mu}_X[n] + 2\hat{\sigma}_X[n] \tag{1}$$

$$s_2[n] = \hat{\mu}_X[n] - 2\hat{\sigma}_X[n] \tag{2}$$

2 Spectral estimation

The file `SpectralEstimation.txt` contains one realization of a random process $X[n]$. Load the signal with the command `X=load('SpectralEstimation.txt')`, and do the following:

1. Plot the signal $X[n]$
2. Compute and plot the basic periodogram of $X[n]$ (the squared FFT of $X[n]$, non-negative frequencies only).
3. Estimate the power spectrum of $X[n]$ with the Welch periodogram, by using the Matlab function `pwelch`. (Read the help to understand how to use it). How many components is the process made of? Change the parameters of the Welch periodogram (length of every section, amount of overlapping, window type) to understand how they influence the estimate of the power spectrum. **Hint:** Set the `NFFT` parameter of the `pwelch` function equal to the number of samples N_{wel} of each section of the signal.